

is automatically saved to the `dconf` key `/apps/gnome-shell / looking-glass-history`, and loaded when you restart the shell.

Slowing down shell animations: You can use Looking Glass to call the special function `St.set_slow_down_factor(factor)` where 'factor' is a value greater than 1 to make all shell animations slower. This is particularly useful when implementing new animation behaviour.

The Picker: The Picker allows you to visually select any object from the shell's scene. When you do so, it is inserted as a result into the Evaluator pane.

Getting started with your first contribution

Now, after building the source code of the particular module, you can start with your first contribution. This process basically consists of going through the source code and finding that part of the code responsible for the bug. For help, you can contact the GNOME developers on the IRC.

Connecting to the GNOME IRC

If your browser is Mozilla Firefox, then you can install the ChatZilla extension for the IRC, which is basically an IRC client developed by Mozilla. If you are not interested in using ChatZilla, you can try IRCCloud, Mibbit or XChat based on what suits you. The next step is to connect to the GNOME IRC server. For this, connect to the GNOME server by using the following command:

```
/server irc.gnome.org
```

Once you are connected to their IRC server, you can join a channel where you can get help for the bug. To join a channel, just type:

```
/join #channel-name
```

After you connect to any channel, you can start asking questions regarding the bug or about anything else too.

Starting with the first bug fix in GNOME

Once you are ready with the build, go to Bugzilla to search for bugs. As a beginner, it is recommended that you start with minor or trivial bugs. Once you find a bug, just mention that you are interested in working on this bug in the 'bug comments' section. After some time, you will get that bug assigned to you. Once you are done with selecting a bug, you can ask for any help regarding the bug in `#gnome-love` on the IRC. After you have fixed the bug, you have to make a patch, for which you have to use Git.

Submitting a patch using Git

Commit a patch: After you've made changes to the code, you are ready for a patch. For this, you need to commit the changes you have made locally.

```
git commit -a
```

In this command, `-a` means all the changes. The `Git commit` command takes you to an editor where you can add the commit message. Your commit message should follow a standard format: a short title, followed by a longer explanation, followed by a link to the relevant bug. Your commit message should look something like this:

Short title describing your changes to the code.

Longer description and explanations if necessary.

Bug link

Create a patch: To create a patch you need to give the following command:

```
git format-patch HEAD^
```

Submit a patch: Send the patch to the project by attaching it to the relevant bug report in GNOME Bugzilla. Unless explicitly stated in the patch file, do not send the patches privately to the maintainer. If your patch is an enhancement and not a bug, you can create a new bug report with *severity -> enhancement*. Don't forget to attach your (Git) patch.

Tips for creating a patch

- In your commit messages, describe both *what* the patch does and *how* it is done, unless the patch is very trivial.
- Strictly follow the coding style used in the code you are working on. This may sound trivial, but consistency is very important. Many projects describe the preferred coding style in the patch file; for other projects, just take a look at the surrounding code and try to use the same format.
- Avoid unrelated changes. This makes the patch easier to read and approve.
- Provide one patch per issue. Do not make big patches that fix unrelated bugs.
- Make sure the patch does not contain trailing whitespace and spurious empty lines. You can use `git diff --check` for this.
- Do not compress the patch with gzip or similar tools, since it makes reviewing more painful. If the patch is big enough to really require compression, chances are that you need to carefully discuss your patch with the maintainer anyway. 

References

- [1] <https://wiki.gnome.org/GnomeLove/SubmittingPatches>
- [2] <https://wiki.gnome.org/action/show/Projects/Jhbuild>

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